UA ATLAS HEP · MSVTX · WORKING DECK

Information theory for ATLAS data

A section-by-section fit pass through Stone, 2nd ed.

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v0 · skeleton, ratings not yet filled

The goal

Define **entropy** and **mutual information** in our own context. Starting from bits did not work.

SPACE 1

Named feature space

The NN1 / NN2 registry inputs. Barrel 20 + 10, endcap 16 + 10. Mixed counts and continuous variables.

SPACE 2

Putative latent space

The 128-dim penultimate activations of each net. Continuous, basis differs per fold.

How each section gets rated

Fit	Meaning	Test
A	Direct fit	We have the data, or already do something like it
B	Adaptable	Needs a bridging assumption or a proxy variable
C	Interesting	No clean ATLAS hook
X	Not applicable	Dropped from the deck
Q	Open question	Must be answered before it can be rated

One slide per section rated A or B. Every rating carries the candidate ATLAS mapping, or the question blocking it.

The book in eleven chapters

Ch	Title	Page	First guess, to challenge
1	What is information?	1	Vocabulary
2	Entropy of discrete variables	21	Core: count features
3	The source coding theorem	45	Mostly background
4	The noisy channel coding theorem	79	Core: 4.2 to 4.4
5	Entropy of continuous variables	113	**Core: MaxEnt, binning bias**
6	Mutual information: continuous	135	**Core: MI, invariance, KL**
7	Channel capacity: continuous	153	7.6 only: MI vs correlation
8	Rate distortion theory	173	Possible: latent bottleneck
9	Transfer entropy	195	Needs a time axis
10	Thermodynamic entropy and information	203	Context only
11	Information as nature's currency	217	Context only

Where the analysis already touches it

BCE LOSS · 2.4, 6.6

BCE is a cross-entropy. Loss below the constant predictor is a lower bound on I(features; label).

dCor PENALTY · 6.4, 7.6

ABCD needs I(NN1; NN2 | background) = 0. dCor is the proxy we train on.

MAXENT BACKGROUND · 5.6

Per-feature MaxEnt marginals, then copula rungs. The gap between rungs is a mutual information.

QUANTILE SCALER · 6.5

MI is invariant under monotone maps. The rank transform changed AUC, not information.

Chapter 1: what is information?

Sec	Title	Pg	Fit	ATLAS mapping or question
1.2	Information, eyes and evolution	2	[ ]	
1.3	Finding a route, bit by bit	3	[ ]	
1.4	A million answers to twenty questions	8	[ ]	
1.5	Information, bits and binary digits	10	[ ]	
1.6	Example 1: telegraphy	11	[ ]	
1.7	Example 2: binary images	13	[ ]	
1.8	Example 3: grey-level image	15	[ ]	

First point on the table: 1.3. Is a preselection cut or an ABCD boundary one binary question about an event?

Open questions

1. Entropy of what, exactly: one feature, the feature vector, the NN score, or the ABCD region label?
2. Which estimator survives 2,280 barrel background events? Binned, kNN, or a variational bound?
3. Differential entropy depends on units and scaling. Do we only ever report differences and MI?
4. Is closure a global MI statement or a local one in the high-high corner?
5. Does tokenizing events give a discrete alphabet where chapter 2 applies directly?